ARCHITECTURE_DETAILED

TrackAGatModel — Detailed Architecture (Quick Review)

Track-A-style parallel model: LSTM (temporal) + self-written multi-head GAT (cross-sectional, `vol->PK` directed edge) + news branch with per-ticker gate, concatenated at the head. Verified against `code/model.py`, `code/gat.py`, `code/run_ablation.py`.

Legend: `B`=batch, `N`=tickers per snapshot, `seq`=sequence length, `price_dim=5`, `news_dim=146`, `hidden=64`, `heads=4`, `gnn_dim = hidden*heads = 256`.

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1. Top-level block diagram (3 parallel branches)

```
INPUTS (one target date's cross-stock panel)
  price     [B, N, seq, 5]      (5 node features; see §4)
  news      [B, N, seq, 146]    (PhoBERT-derived news manifest)
  news_mask [B, N, seq]
  ticker_ids[B, N]
  adjacency [B, N, N]           (directed vol->PK Top-5, self-loops, frozen
TRAIN-only)

 ┌──────────────────────────── BRANCH 1: PRICE-LSTM (temporal, per node)
────────────────────────────┐
 │ price [B,N,seq,5]
│
 │   -> reshape [B*N, seq, 5]
│
 │   -> LSTM(in=5, hidden=64, layers=2, dropout=0.2)   -> take last step
│
 │   -> h_lstm  [B, N, 64]
│

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 ┌─────────────────────── BRANCH 2: GAT graph (cross-sectional, RAW feats @ t)
───────────────────────┐
 │ node_raw = price[:, :, -1, :]        # LAST timestep RAW features, NOT
h_lstm                │
```

```
 │           [B, N, 5]
│
 │   -> GATLayer gat1 (5 -> 64, heads=4; concat)   [B, N, 256]
│
 │   -> GATLayer gat2 (256 -> 64, heads=4; concat) [B, N, 256] = h_gnn
│
 │      attention masked by (adjacency>0), softmax over SOURCE nodes,
LeakyReLU + ELU           │
 │      graph OFF  => adjacency := identity (self-loops only)
│

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 ┌─────────────────────── BRANCH 3: NEWS (per node) + per-ticker GATE
────────────────────────┐
 │ news * news_mask  [B,N,seq,146]
│
 │   -> Linear(146 -> 64) + ReLU
│
 │   -> LSTM(64 -> 64, layers=2, dropout=0.2) -> last step -> news_hidden
[B,N,64]                │
 │   -> gate = sigmoid(gate_logits[ticker_ids])         # scalar per ticker,
in (0,1)             │
 │   -> gated_news = gate * news_hidden   [B, N, 64]
│

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                    h_lstm[64]        h_gnn[256]        gated_news[64]
                        └──────────────────┼─────────────────────┘
                                           v
                       HEAD:  concat -> h  [B, N, 384]
                              Linear(384 -> 64) + ReLU + Dropout(0.2)
                              Linear(64 -> 1)              -> raw  [B, N]
                                           v
                    POSITIVITY FLOOR (denormalized space, eps=1e-6)
                       denorm  = raw*std + mean                 (per-ticker
StandardScaler)
                       floored = eps*softplus(denorm/eps) + eps (>= 0; target
is Parkinson VARIANCE)
                       out     = (floored - mean) / std         [B, N]
(normalized space)
```

Notes: - Branch 2 consumes RAW node features at the last timestep, not the LSTM hidden state. This keeps the graph branch an independent cross-sectional view (fair leave-one-out graph ablation) and matches the `vol->PK` edge semantics (`volume_shock_i(t) -> sqrt(PK_j)(t+1)`). - Without news (`use_news=False`), `gated_news` is zeros `[B,N,64]`; the gate is a no-op.

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2. Directed `vol->PK` Top-5 edge

Adjacency built by lead-lag correlation, directed and asymmetric, frozen on TRAIN only.

```
For each TARGET ticker j, connect the Top-5 SOURCE tickers i ranked by:
    corr( volume_shock_i(t) ,  sqrt(PK_j)(t+1) )       # source volume LEADS
target volatility

        volume_shock          sqrt(PK) next day
   [i=SRC] ──────────────────────────────────> [j=DST]
    (leads)        directed, asymmetric          (lags)

   adjacency[i, j] = 1  for i in Top5(j);   diagonal self-loops kept
(adjacency[j, j] = 1)

Attention (per GAT layer): mask entries where adjacency<=0, softmax over
SOURCE j, aggregate.
```

Chosen over undirected k-NN correlation-on-PK because: - (a) lead-lag is predictive/causal, not contemporaneous - (b) volume leads volatility - (c) directed captures who-leads-whom - (d) EDA-selected Top-5 (not an arbitrary k)

Leakage safety: edges are computed on TRAIN data only and frozen for val/test.

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3. Leave-one-out ablation (`run_ablation.py`, h in {1, 5, 10, 22})

Build FULL, then RETRAIN each variant from scratch with exactly ONE component removed, so every effect is measured on the same footing (each variant trains in the same graph on/off regime it is evaluated in — no train/eval mismatch).

```
                         FULL (LSTM + vol->PK GAT graph + news + per-ticker
gate)
                                  |
        ┌─────────────────────────┼────────────────────┬───────────────────────┐
        v                         v                    v                       v
   minus_graph        minus_gate          minus_news               (external)
   DROP whole GAT     drop per-ticker     drop news branch           HAR
   branch (no         gate                (gate no-op w/o news)    pooled
linear reg
   node/edge/GAT)     retrained           retrained                  fit
   retrained
```

Variant	LSTM	Graph (vol->PK)	News	Gate	Trained
HAR	-	-	-	-	pooled linear (ext)
FULL	yes	yes	yes	yes	retrained
minus_graph	yes	NONE (no GAT)	yes	yes	retrained
minus_gate	yes	yes	yes	OFF	retrained

Variant	LSTM	Graph (vol->PK)	News	Gate	Trained
minus_news	yes	yes	OFF	n/a	retrained

Contribution of component X (test QLIKE):

```
effect(X) = QLIKE(FULL) - QLIKE(minus_X)
    negative  => removing X hurt (QLIKE rose)  => X HELPED
    positive  => removing X improved QLIKE     => X did not help
```

Significance: Diebold-Mariano (HLN correction) on seed-ensembled test predictions, for FULL vs each minus_X and FULL vs HAR.

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4. The 5 node features (shared by the LSTM and GAT branches)

Column order: `[pk_daily, har_weekly, har_monthly, market_pk, volume_zscore(22)]`.

#	Feature	Definition	Causal/leakage
1	pk_daily	`parkinson_volatility` (the variance), clipped ±3σ (train stats)	uses day `t` only
2	har_weekly	rolling-5 mean of pk	trailing
3	har_monthly	rolling-22 mean of pk	trailing
4	market_pk	cross-sectional MEDIAN of `sqrt(PK)` across present tickers at `t` (market factor)	contemporaneous (col `t`)
5	volume_zscore	`(log1p(volume) - roll.mean) / roll.std`, **rolling-22** window; no-volume tickers → 0.0	trailing

Feature 5 uses a **22-trading-day (1-month) window**, unified with `har_monthly`'s rolling(22) (TrackA-local override in `run_trackA.build_trackA_basis`; the underlying column key retains the legacy name `volume_zscore_20` from the 2026-08-11 EDA baseline, which stays at window 20 so its recorded results remain reproducible). SEQ (LSTM lookback) = 22. The GAT node input is the last-timestep feature vector, which already encodes multi-scale rolling context via features 2/3/5. ```